

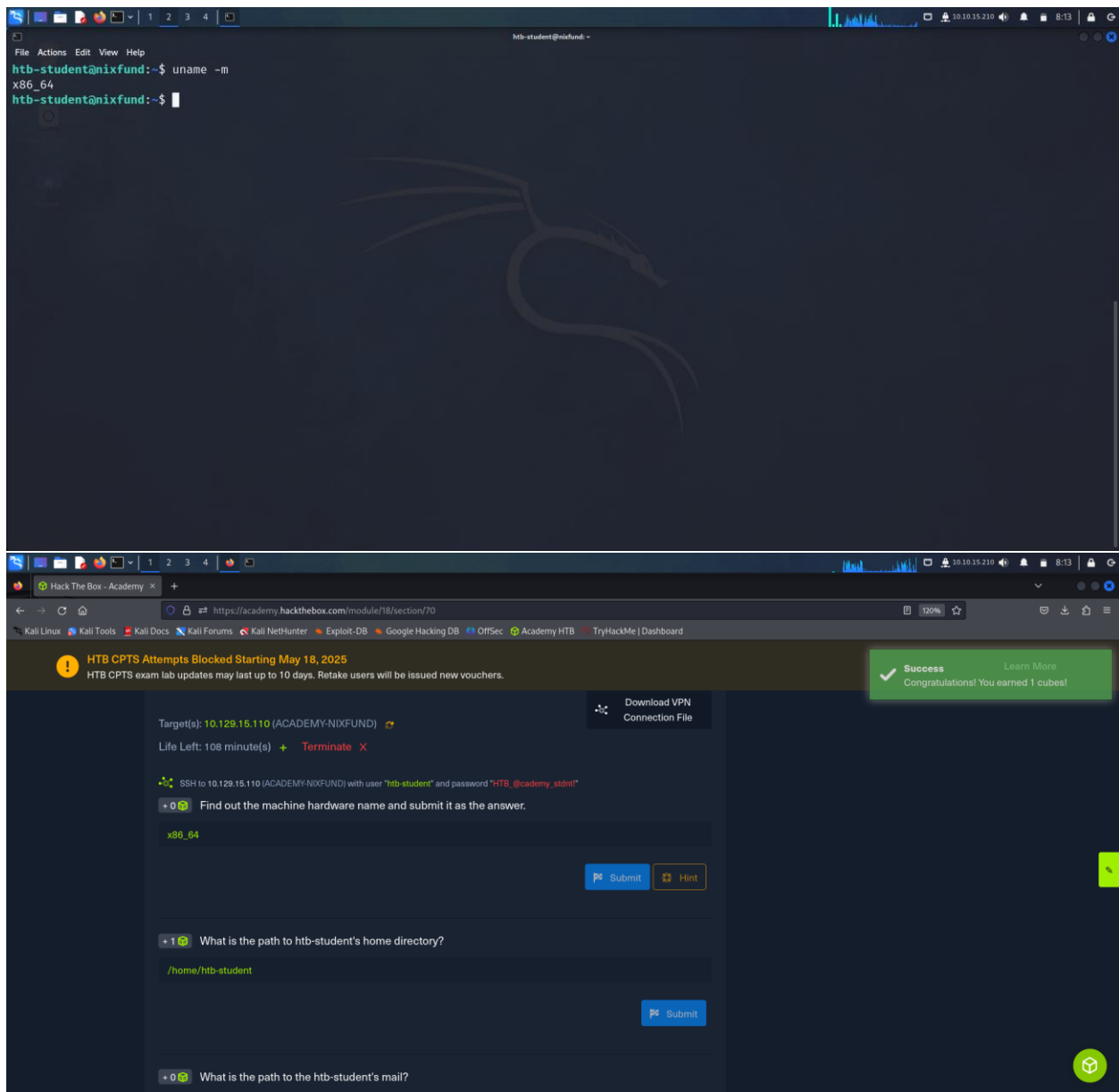
1. Introduction

Learning Linux fundamentals has been an exciting journey for me, opening up a whole new world of possibilities with an operating system that powers so much of today's technology, from servers to embedded devices. By diving into hands-on exercises through Hack The Box Academy, I've gained practical skills in Linux commands, system administration, and network management. For instance, I learned how to start a simple web server using commands like `"http-server -p 8080"` and `"php -S 127.0.0.1:8080"`, which gave me a sense of control over network services. I also explored system details, like discovering the machine hardware name `"x86_64"`, and navigated the file system to find paths such as `"/home/htb-student"` for a user's home directory and `"/var/mail/htb-student"` for their mail. Additionally, I tackled tasks like determining the number of partitions in my Pwnbox, which turned out to be `"3"`, helping me understand how to manage storage and resources better. These experiences have been crucial in building my foundational skills and preparing me to handle real-world Linux environments.

2. Module Sections

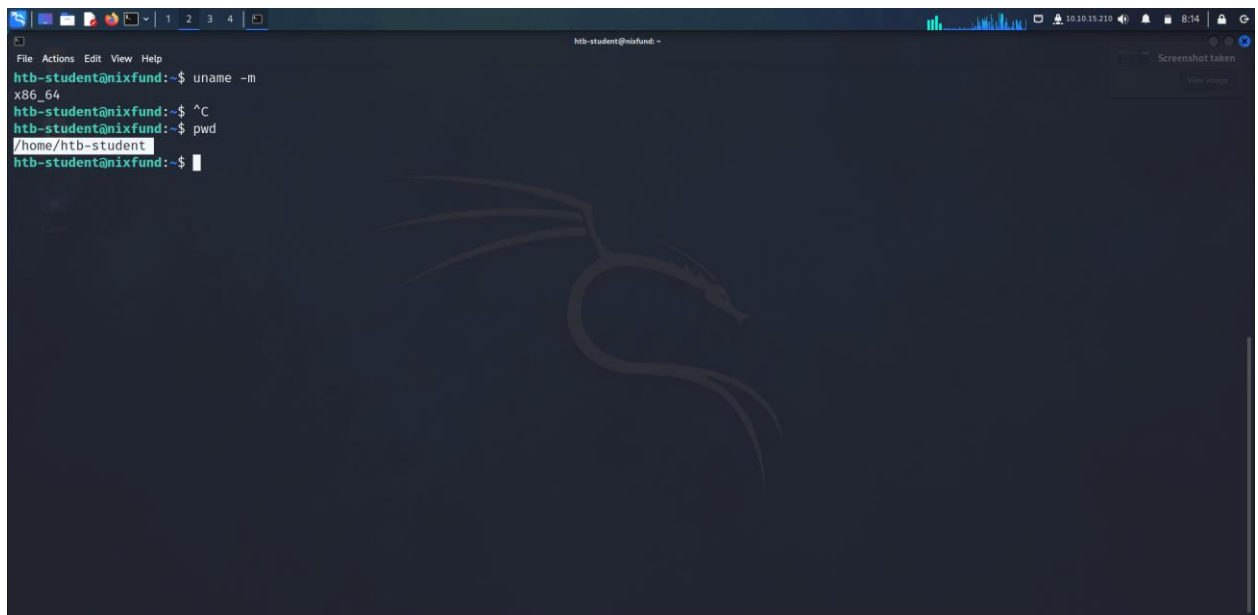
Question: Find out the machine hardware name and submit it as the answer. Answer: `x86_64`

I used the command `uname -m` to get the hardware name of the machine



Question: What is the path to htb-student's home directory? Answer: /home/htb-student

I printed the current working directory using pwd to get the users home directory

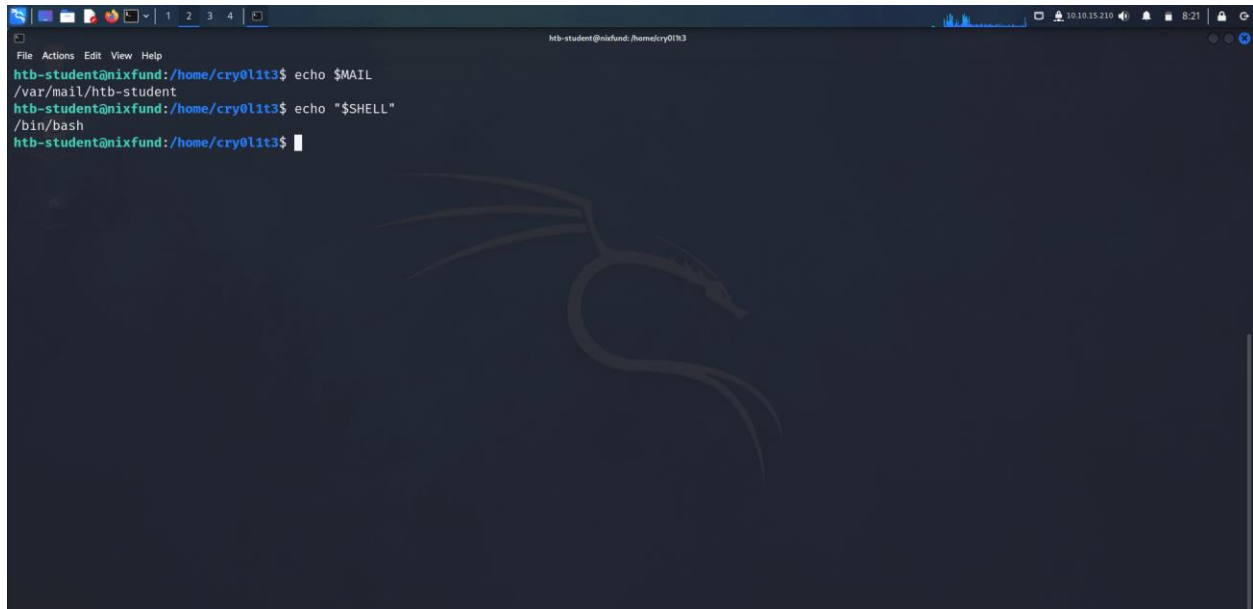


Question: What is the path to the htb-student's mail directory? Answer: /var/mail/htb-student

First, I typed `echo $MAIL` and pressed Enter. When I ran this command, it showed me the value of the MAIL environment variable, which Linux had set to point to my default mail directory.

Question: Which shell is specified for the htb-student user? Answer: /bin/bash

Then, I typed `echo "$SHELL"` and hit Enter. This command displayed the full path to the shell I was using, stored in the SHELL environment variable. I got /bin/bash as the output, which meant I was using the Bash shell at the time.

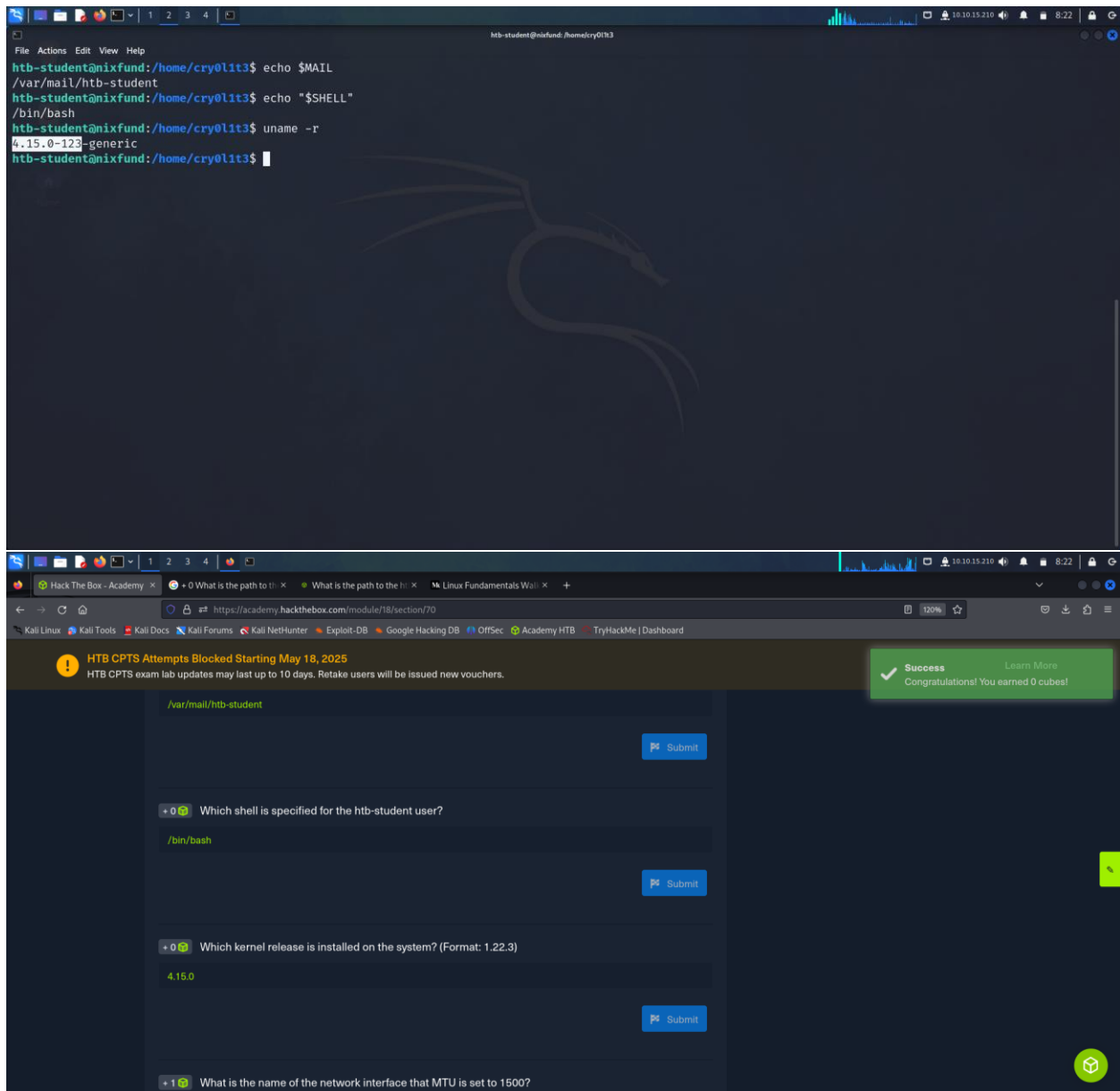
A screenshot of a terminal window with a dark blue background and a faint dragon logo. The terminal shows the following commands and outputs:

```
htb-student@nixfund:/home/cry01t3$ echo $MAIL
/var/mail/htb-student
htb-student@nixfund:/home/cry01t3$ echo "$SHELL"
/bin/bash
htb-student@nixfund:/home/cry01t3$
```

The terminal window has a menu bar with 'File', 'Actions', 'Edit', 'View', and 'Help'. The title bar shows 'htb-student@nixfund:/home/cry01t3'.

Question: Which kernel release is installed on the system? (Format: 1.2.3) Answer: 4.15.0

I used the uname with the -r tag to specify the details of the kernel release



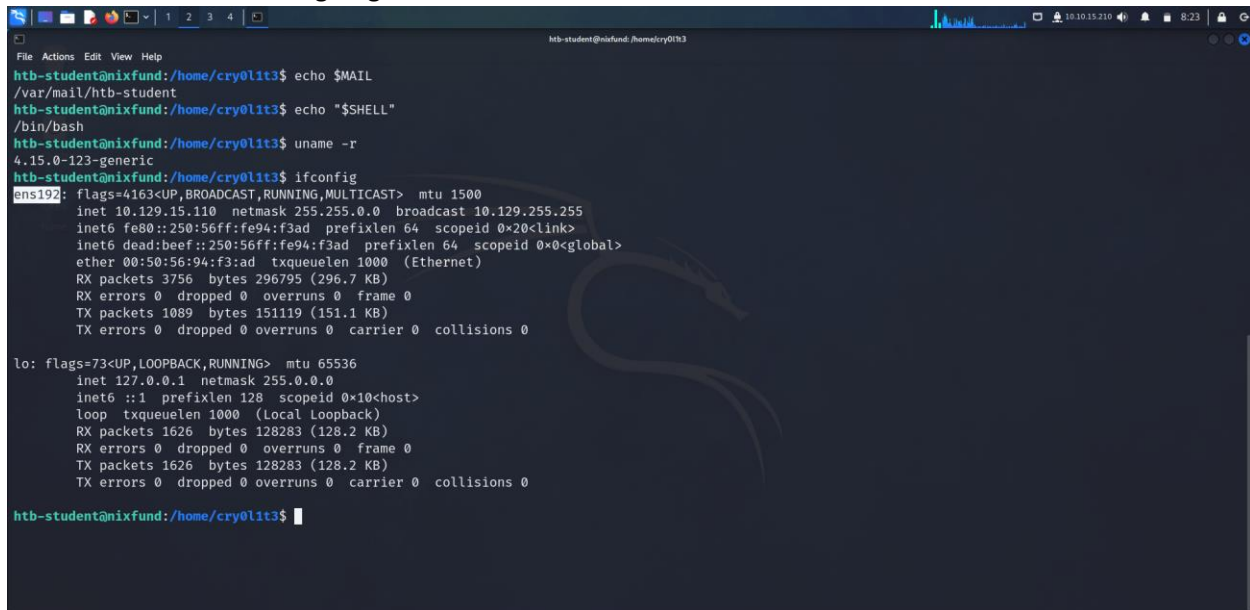
The image shows a terminal window and a web browser. The terminal window displays the following commands and output:

```
htb-student@nixfund:/home/cry01t3$ echo $MAIL
/var/mail/htb-student
htb-student@nixfund:/home/cry01t3$ echo "$SHELL"
/bin/bash
htb-student@nixfund:/home/cry01t3$ uname -r
4.15.0-123-generic
htb-student@nixfund:/home/cry01t3$
```

The web browser shows the Hack The Box Academy interface. A banner at the top reads: "HTB CPTS Attempts Blocked Starting May 18, 2025. HTB CPTS exam lab updates may last up to 10 days. Retake users will be issued new vouchers." A green success message in the top right corner says: "Success! Congratulations! You earned 0 cubes!" The main content area shows a list of challenges. The third challenge is selected, with the question: "Which kernel release is installed on the system? (Format: 1.22.3)". The answer "4.15.0" is entered in the text box. Below the answer is a "Submit" button. The first challenge is partially visible, asking for the path to the htb-student user's mail, with the answer "/var/mail/htb-student" entered. The second challenge asks for the shell specified for the htb-student user, with the answer "/bin/bash" entered. The fourth challenge is partially visible, asking for the name of the network interface that MTU is set to 1500.

Question: What is the name of the network interface that MTU is set to 1500? Answer: ens192

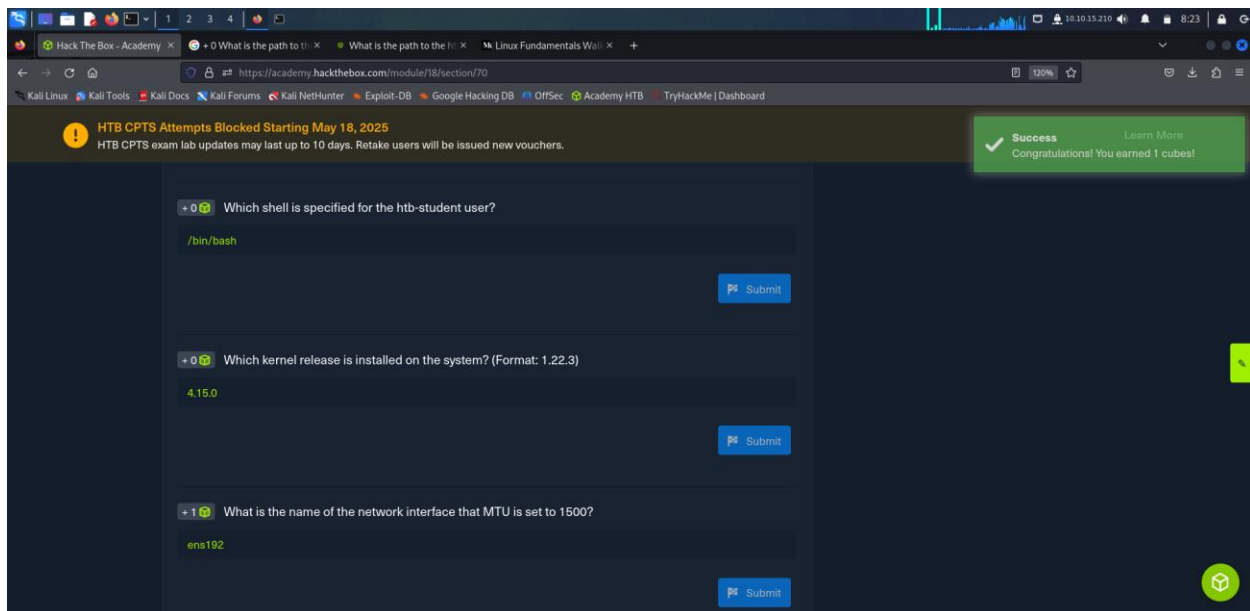
I ran the command `ifconfig` to get the network details and saw MTU was set to 1500



```
htb-student@nixfund:/home/cry01t3$ echo $MAIL
/var/mail/htb-student
htb-student@nixfund:/home/cry01t3$ echo "$SHELL"
/bin/bash
htb-student@nixfund:/home/cry01t3$ uname -r
4.15.0-123-generic
htb-student@nixfund:/home/cry01t3$ ifconfig
ens192: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.129.15.110 netmask 255.255.0.0 broadcast 10.129.255.255
    inet6 fe80::250:56ff:fe94:f3ad prefixlen 64 scopeid 0x20<link>
    inet6 dead:beef::250:56ff:fe94:f3ad prefixlen 64 scopeid 0x0<global>
    ether 00:50:56:94:f3:ad txqueuelen 1000 (Ethernet)
    RX packets 3756 bytes 296795 (296.7 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1089 bytes 151119 (151.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1626 bytes 128283 (128.2 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1626 bytes 128283 (128.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

htb-student@nixfund:/home/cry01t3$
```



Question: What is the name of the hidden ".history" file in the htB-user's home directory? Answer:

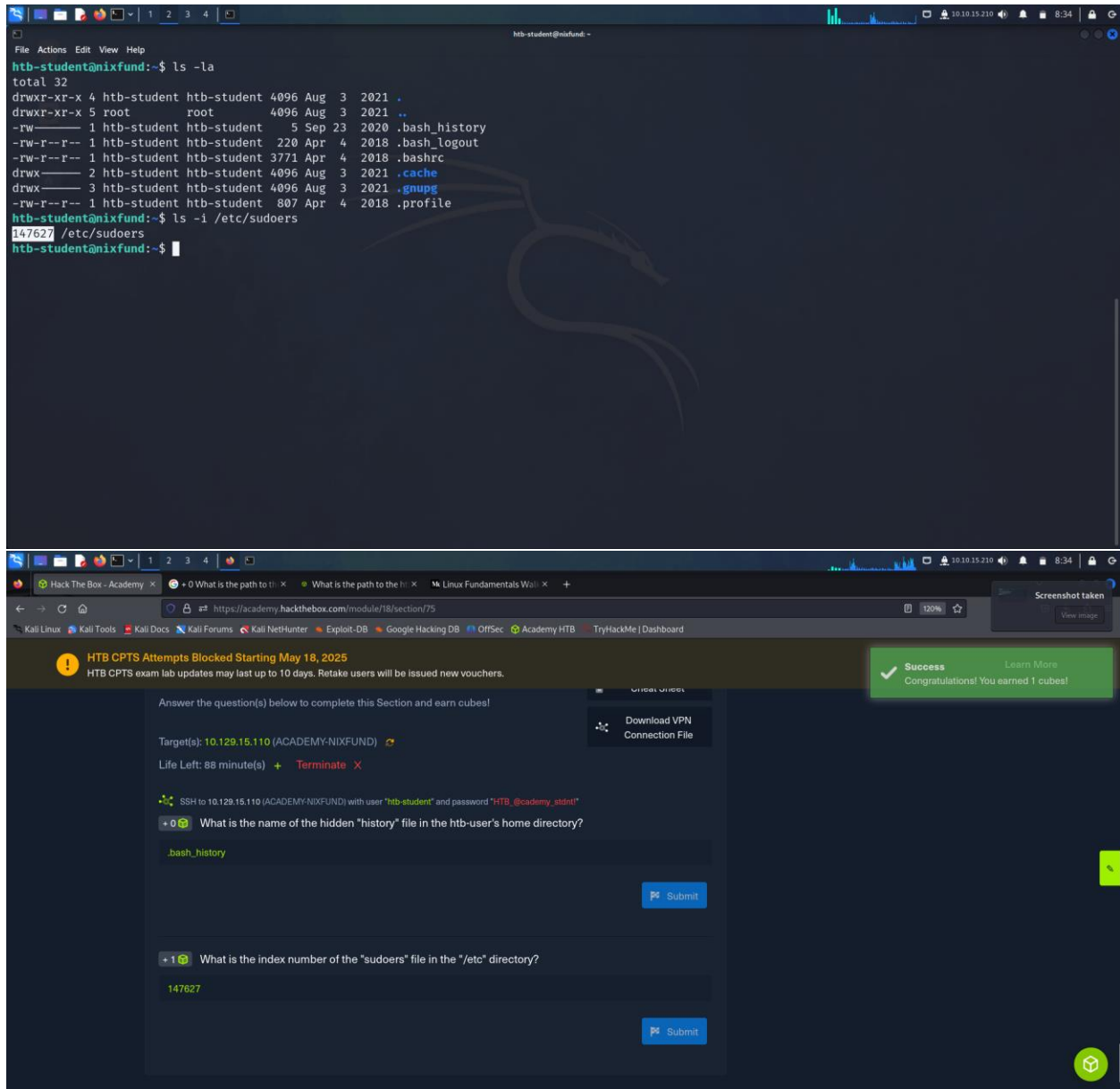
`.bash_history`

I used `ls -la` to list all files and directories and saw `.bash` history was hidden

```
htb-student@nixfund:~$ ls -la
total 32
drwxr-xr-x 4 htb-student htb-student 4096 Aug 3 2021 .
drwxr-xr-x 5 root         root         4096 Aug 3 2021 ..
-rw-r--r-- 1 htb-student htb-student  5 Sep 23 2020 .bash_history
-rw-r--r-- 1 htb-student htb-student 220 Apr 4 2018 .bash_logout
-rw-r--r-- 1 htb-student htb-student 3771 Apr 4 2018 .bashrc
drwxr-xr-x 2 htb-student htb-student 4096 Aug 3 2021 .cache
drwxr-xr-x 3 htb-student htb-student 4096 Aug 3 2021 .gnupg
-rw-r--r-- 1 htb-student htb-student 807 Apr 4 2018 .profile
htb-student@nixfund:~$
```

Question: What is the index number of the "sudoers" file in the "/etc" directory? Answer: 147627

To examine the inode number of the /etc/shadow file, I used the command `ls -li /etc/shadow` as shown below



Question: What is the name of the last modified file in the "/var/backups" directory? Answer: apt.extended_states.0

To explore the detailed and time-sorted listing of files in my current directory, I used the command `ls -lt`

```
htb-student@nixfund: /var/backups
File Actions Edit View Help
/var/backups/dpkg.status.4.gz
/var/backups/dpkg.status.5.gz
/var/backups/dpkg.status.6.gz
/var/backups/group.bak
/var/backups/gshadow.bak
/var/backups/passwd.bak
/var/backups/shadow.bak
htb-student@nixfund:~$ cd /var/backups
htb-student@nixfund:/var/backups$ ls -la -t
total 2168
drwxr-xr-x  2 root root    4096 Aug  3  2021 .
-rw-r--r--  1 root root   41872 Nov 12  2020 apt.extended_states.0
-rw-r--r--  1 root root    4437 Nov 12  2020 apt.extended_states.1.gz
-rw-r--r--  1 root root    742750 Nov 11  2020 dpkg.status.0
-rw-r--r--  1 root root    206270 Nov 11  2020 dpkg.status.1.gz
-rw-r--r--  1 root root    206270 Nov  5  2020 dpkg.status.2.gz
-rw-r--r--  1 root root    206270 Nov  5  2020 dpkg.status.3.gz
-rw-r--r--  1 root root    206270 Nov  5  2020 dpkg.status.4.gz
-rw-r--r--  1 root root    206270 Nov  5  2020 dpkg.status.5.gz
-rw-r--r--  1 root root    206270 Nov  5  2020 dpkg.status.6.gz
-rw-r--r--  1 root root   51200 Oct 29  2020 alternatives.tar.0
-rw-r--r--  1 root root    4623 Oct 22  2020 apt.extended_states.2.gz
-rw-r--r--  1 root root    2497 Oct 16  2020 alternatives.tar.1.gz
-rw-r--r--  1 root root    4601 Oct 15  2020 apt.extended_states.3.gz
-rw-r--r--  1 root root    2492 Sep 24  2020 alternatives.tar.2.gz
-rw-r--r--  1 root root     367 Sep 23  2020 dpkg.statoverride.0
-rw-r--r--  1 root root     229 Sep 23  2020 dpkg.statoverride.1.gz
-rw-r--r--  1 root root     229 Sep 23  2020 dpkg.statoverride.2.gz
-rw-r--r--  1 root root     229 Sep 23  2020 dpkg.statoverride.3.gz
-rw-r--r--  1 root root     229 Sep 23  2020 dpkg.statoverride.4.gz
-rw-r--r--  1 root root     229 Sep 23  2020 dpkg.statoverride.5.gz
```

Question: What is the inode number of the "shadow.bak" file in the "/var/backups" directory? Answer: 265293

I opened my terminal and typed `ls -i | grep shadow.bak`, then pressed Enter. The `ls -i` command listed the files in my current directory along with their inode numbers, which are unique identifiers for each file on the filesystem. I piped this output to `grep shadow.bak`, which filtered the results to show only the line containing `shadow.bak`

```
htb-student@nixfund: /var/backups
File Actions Edit View Help
htb-student@nixfund:/var/backups$ ls -i
262248 alternatives.tar.0      262233 apt.extended_states.4.gz  262235 dpkg.diversions.0.gz  262236 dpkg.statoverride.0.gz  262230 dpkg.status.0.gz
262559 alternatives.tar.1.gz  262178 dpkg.diversions.0    262231 dpkg.statoverride.0    263999 dpkg.status.0        265226 group.bak
262261 alternatives.tar.2.gz  262203 dpkg.diversions.1.gz     262205 dpkg.statoverride.1.gz  262179 dpkg.status.1.gz     265817 gshadow.bak
266334 apt.extended_states.0  262264 dpkg.diversions.2.gz     262310 dpkg.statoverride.2.gz  262234 dpkg.status.2.gz     264599 passwd.bak
266335 apt.extended_states.1.gz 262257 dpkg.diversions.3.gz 262311 dpkg.statoverride.3.gz 262241 dpkg.status.3.gz     265293 shadow.bak
266430 apt.extended_states.2.gz 262246 dpkg.diversions.4.gz 262247 dpkg.statoverride.4.gz 262243 dpkg.status.4.gz
264827 apt.extended_states.3.gz 262249 dpkg.diversions.5.gz 262250 dpkg.statoverride.5.gz 262220 dpkg.status.5.gz
htb-student@nixfund:/var/backups$ ls -l | grep shadow.bak
gshadow.bak
shadow.bak
htb-student@nixfund:/var/backups$ ls -i | grep shadow.bak
265817 gshadow.bak
265293 shadow.bak
htb-student@nixfund:/var/backups$
```

HTB CPTS Attempts Blocked Starting May 18, 2025
HTB CPTS exam lab updates may last up to 10 days. Retake users will be issued new vouchers.

Success
Congratulations! You earned 1 cubes!

Learn More

Questions

Answer the question(s) below to complete this Section and earn cubes!

Target(s): 10.129.15.110 (ACADEMY-NIXFUND) 🏠

Life Left: 80 minute(s) ➕ Terminate ✕

SSH to 10.129.15.110 (ACADEMY-NIXFUND) with user "htb-student" and password "HTB_academy_stdnt!"

+ 0 🏠 What is the name of the last modified file in the "/var/backups" directory?

apt.extended_states.0

Submit

+ 1 🏠 What is the inode number of the "shadow.bak" file in the "/var/backups" directory?

265293

Submit

Cheat Sheet

Download VPN Connection File

🏠

🏠

Question: What is the name of the config file that has been created after 2020-03-03 and is smaller than 28k but larger than 25k? Answer: 00-mesa-defaults.conf

I used the find command with the the set tags to find this

Question: How many files exist on the system that have the ".bak" extension? Answer: 4

I used locate command to get all the files with .bak as their extension

Question: Submit the full path of the "xxd" binary. Answer: /usr/bin/xxd

To determine the full path of the xxd command on my system, I used the which xxd command

```
htb-student@nixfund: /var/backups
File Actions Edit View Help
htb-student@nixfund: /var/backups$ find / -iname "*.conf" -size +25k -size -28k -newermt 2020-03-03 2>/dev/null
/usr/share/dirc.d/00-mesa-defaults.conf
htb-student@nixfund: /var/backups$ locate *.bak
/var/backups/group.bak
/var/backups/gshadow.bak
/var/backups/passwd.bak
/var/backups/shadow.bak
htb-student@nixfund: /var/backups$ which xxd
/usr/bin/xxd
htb-student@nixfund: /var/backups$
```

HTB CPTS Attempts Blocked Starting May 18, 2025
HTB CPTS exam lab updates may last up to 10 days. Retake users will be issued new vouchers.

Success
Congratulations! You earned 0 cubes!

SSH to 10.129.15.110 (ACADEMY-NIXFUND) with user "htb-student" and password "HTB_academy_stdnt!"

+1 🧐 What is the name of the config file that has been created after 2020-03-03 and is smaller than 28k but larger than 25k?

00-mesa-defaults.conf

Submit

+1 🧐 How many files exist on the system that have the ".bak" extension?

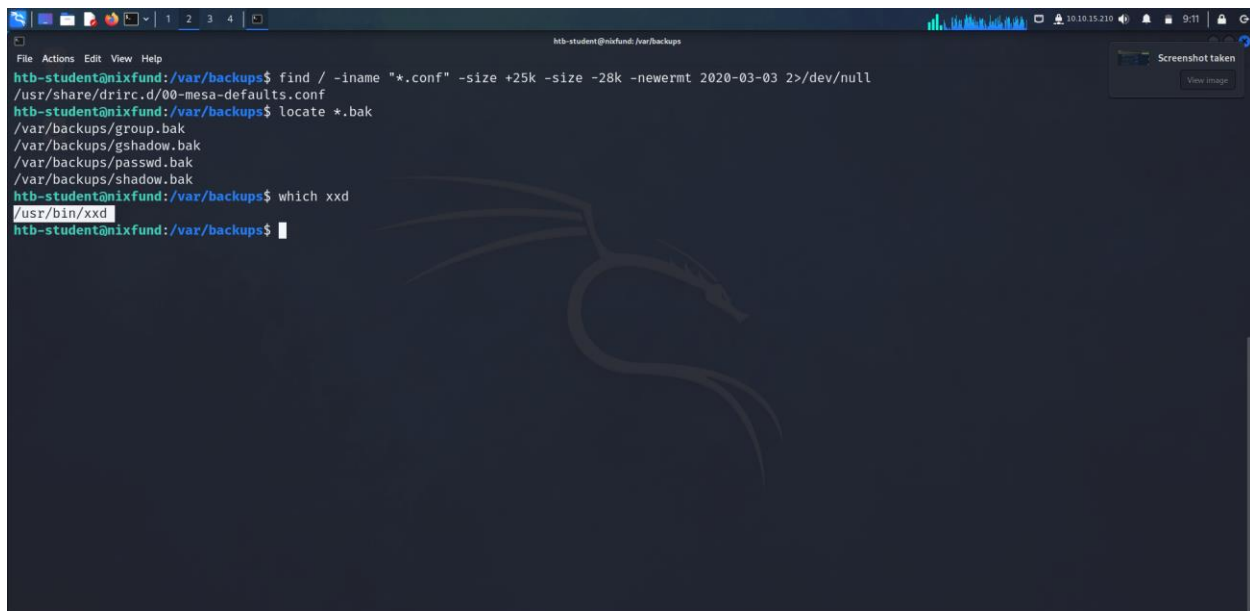
4

Submit

+0 🧐 Submit the full path of the "xxd" binary.

/usr/bin/xxd

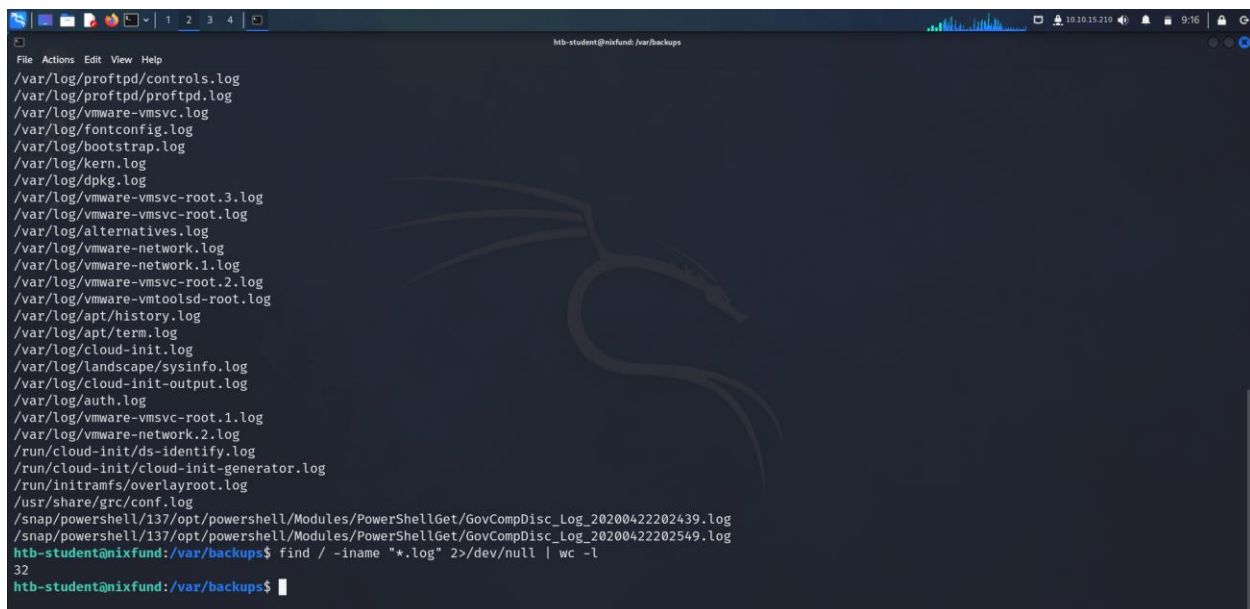
Submit

A screenshot of a Linux terminal window with a dark background and a dragon logo. The terminal shows a series of commands and their outputs. The first command is a find command to search for .conf files. The second is a locate command for .bak files. The third is a which command for xxd. The terminal window has a menu bar with 'File', 'Actions', 'Edit', 'View', and 'Help'. The title bar shows 'htb-student@nixfund: /var/backups'. The status bar at the bottom shows the time as 9:11 and a battery level indicator.

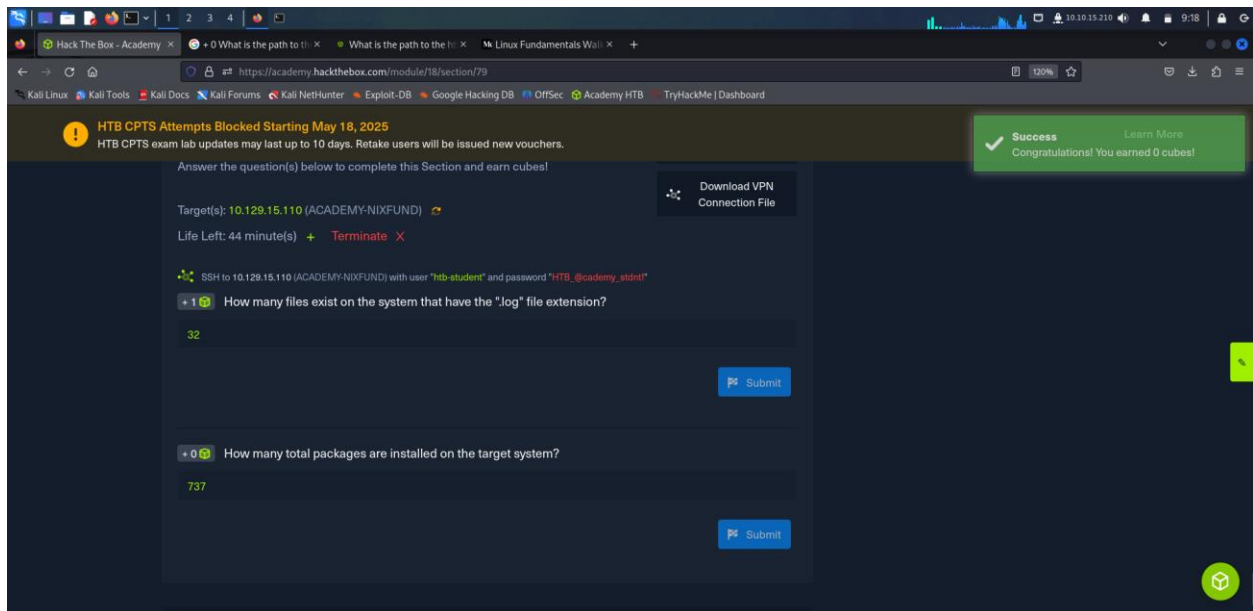
```
htb-student@nixfund:/var/backups$ find / -iname "*.conf" -size +25k -size -28k -newermt 2020-03-03 2>/dev/null
/usr/share/dirc.d/00-mesa-defaults.conf
htb-student@nixfund:/var/backups$ locate *.bak
/var/backups/group.bak
/var/backups/gshadow.bak
/var/backups/passwd.bak
/var/backups/shadow.bak
htb-student@nixfund:/var/backups$ which xxd
/usr/bin/xxd
htb-student@nixfund:/var/backups$
```

Question: How many files exist on the system that have the ".log" file extension? Answer: 32

To count the number of files with a .log extension on my system while suppressing error messages, I used the command `find / -iname "*.log" 2>/dev/null | wc -l` in the Linux terminal

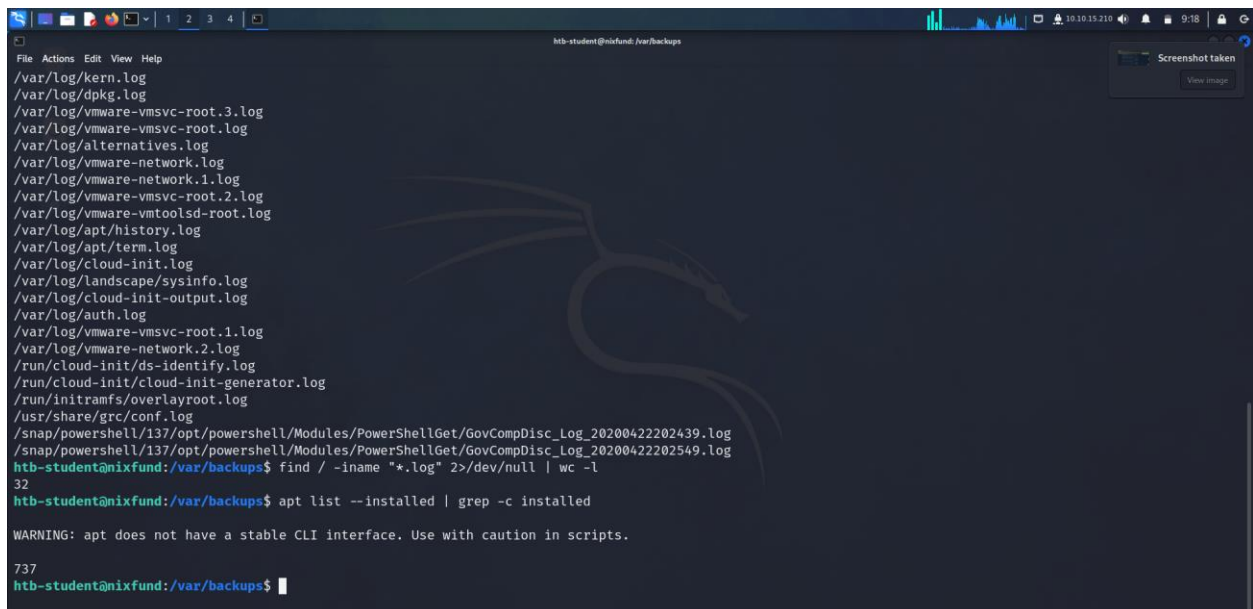
A screenshot of a Linux terminal window with a dark background and a dragon logo. The terminal shows the output of a find command for .log files. The output lists various log files in the /var/log directory and its subdirectories, as well as some files in /run and /usr/share. The final line shows the command being used to count the files, resulting in the number 32. The terminal window has a menu bar with 'File', 'Actions', 'Edit', 'View', and 'Help'. The title bar shows 'htb-student@nixfund: /var/backups'. The status bar at the bottom shows the time as 9:16 and a battery level indicator.

```
/var/log/proftpd/controls.log
/var/log/proftpd/proftpd.log
/var/log/vmware-vmtoolsd.log
/var/log/fontconfig.log
/var/log/bootstrap.log
/var/log/kern.log
/var/log/dpkg.log
/var/log/vmware-vmtoolsd-root.3.log
/var/log/vmware-vmtoolsd-root.log
/var/log/alternatives.log
/var/log/vmware-network.log
/var/log/vmware-network.1.log
/var/log/vmware-vmtoolsd-root.2.log
/var/log/vmware-vmtoolsd-root.log
/var/log/apt/history.log
/var/log/apt/term.log
/var/log/cloud-init.log
/var/log/landscape/sysinfo.log
/var/log/cloud-init-output.log
/var/log/auth.log
/var/log/vmware-vmtoolsd-root.1.log
/var/log/vmware-network.2.log
/run/cloud-init/ds-identify.log
/run/cloud-init/cloud-init-generator.log
/run/initramfs/overlayroot.log
/usr/share/grc/conf.log
/snap/powershell/137/opt/powershell/Modules/PowerShellGet/GovCompDisc_Log_20200422202439.log
/snap/powershell/137/opt/powershell/Modules/PowerShellGet/GovCompDisc_Log_20200422202549.log
htb-student@nixfund:/var/backups$ find / -iname "*.log" 2>/dev/null | wc -l
32
htb-student@nixfund:/var/backups$
```



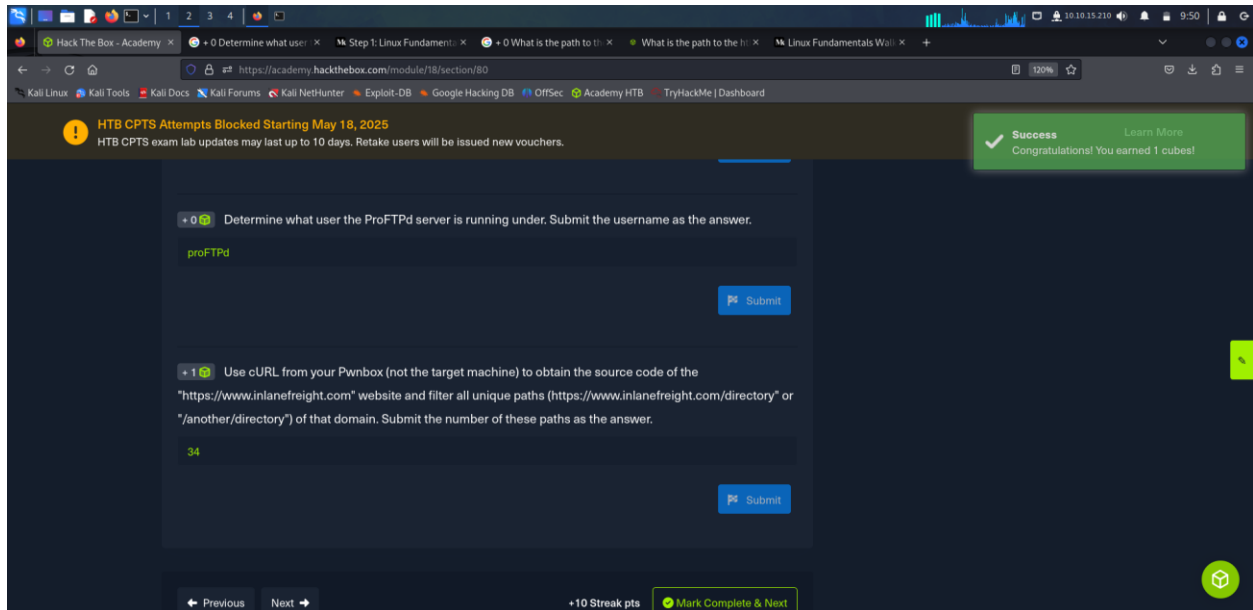
Question: How many total packages are installed on the target system? Answer: 737

To count the number of installed packages on my system, I used the command `apt list --installed | grep -c installed`. I opened my terminal and typed `apt list --installed | grep -c installed`, then pressed Enter. The `apt list --installed` command generated a list of all installed packages on my system, with each line typically ending in `[installed]` to indicate the package status. I piped this output to `grep -c installed`, where `grep -c` counted the number of lines containing the word "installed"



Question: Determine what user the ProFTPD server is running under. Submit the username as the answer. Answer: proftpd

Question: Use cURL from your Pwnbox (not the target machine) to obtain the source code of the "https://www.inlanefreight.com" website and filter all unique paths (https://www.inlanefreight.com/directory" or "/another/directory/") of that domain. Submit the number of these paths as the answer. Answer: 34



Question: Which option needs to be set to create a home directory for a new user using "useradd" command? Answer: -m

Question: Which option needs to be set to lock a user account using the "usermod" command? (long version of the option) Answer: --lock

Question: Which option needs to be set to execute a command as a different user using the "su" command? (long version of the option) Answer: --command

The screenshot shows a web browser window with the URL <https://academy.hackthebox.com/module/18/section/71>. The page has a dark theme and a navigation bar at the top with links to various resources. A yellow banner at the top of the main content area reads: "HTB CPTS Attempts Blocked Starting May 18, 2025. HTB CPTS exam lab updates may last up to 10 days. Retake users will be issued new vouchers." Below this, the "Questions" section is titled "Answer the question(s) below to complete this Section and earn cubes!". There are three questions, each with a "+0" score and a "Submit" button. The first question asks for the option to create a home directory for a new user using "useradd", with the answer "-m" entered. The second question asks for the long version of the option to lock a user account using "usermod", with the answer "--lock" entered. The third question asks for the long version of the option to execute a command as a different user using "su", with the answer "--command" entered. A "Cheat Sheet" button is visible next to the first question. On the right side of the page, there are two green buttons: a "Learn More" button and a "Submit" button.

HTB CPTS Attempts Blocked Starting May 18, 2025
HTB CPTS exam lab updates may last up to 10 days. Retake users will be issued new vouchers.

Learn More

Questions

Answer the question(s) below to complete this Section and earn cubes!

+0 Which option needs to be set to create a home directory for a new user using "useradd" command?

-m

Submit

+0 Which option needs to be set to lock a user account using the "usermod" command? (long version of the option)

--lock

Submit

+0 Which option needs to be set to execute a command as a different user using the "su" command? (long version of the option)

--command

Submit

Question: Use the "systemctl" command to list all units of services and submit the unit name with the description "Load AppArmor profiles managed internally by snapd" as the answer. Answer:
snapd.apparmor.service

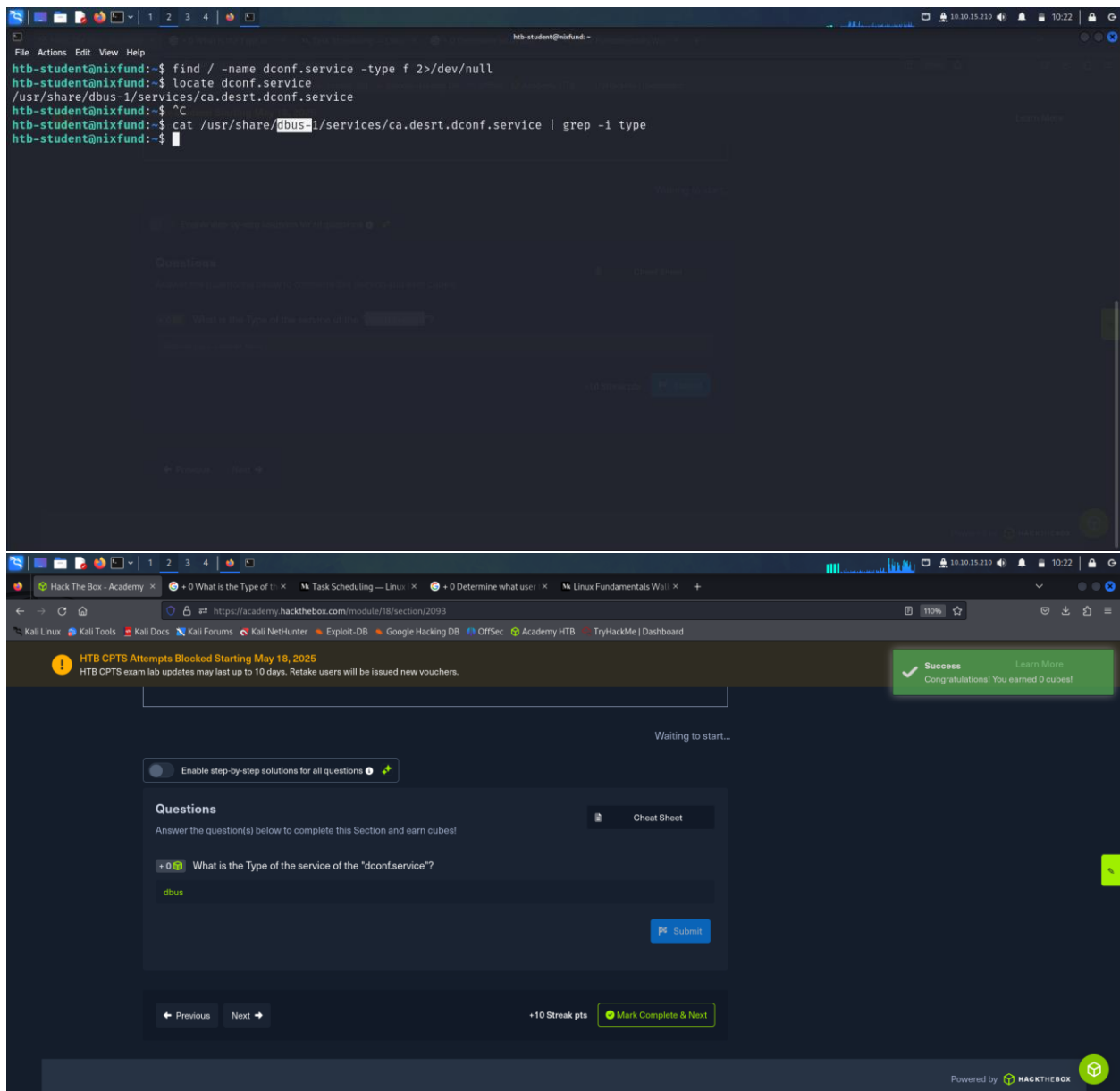
I opened my terminal and typed `systemctl | grep load`, then pressed Enter. The `systemctl` command listed all the units managed by the `systemd` init system. I piped this output to `grep load`, which filtered the results to show only the lines containing the word "load".

The image shows a terminal window and a web browser side-by-side. The terminal window is running the command `systemctl | grep Load` on a system with user `htb-student` at host `nixfund`. The output lists several services and their states:

Service	State	Load	
<code>snapped.apparmor.service</code>	loaded active	exited	Load AppArmor profiles managed internally by snapd
<code>systemd-modules-load.service</code>	loaded active	exited	Load Kernel Modules
<code>systemd-random-seed.service</code>	loaded active	exited	Load/Save Random Seed
<code>systemd-rfkill.socket</code>	loaded active	listening	Load/Save RF Kill Switch S

The web browser shows the HackTheBox interface for a challenge titled "Determine what user?". The challenge description asks the user to use the `systemctl` command to list all units of services and submit the unit name with the description "AppArmor profiles managed internally by snapd" as the answer. The user has successfully completed the challenge and earned 1 cube.

Question: What is the Type of the service of the "dconf" user? Answer: dbus

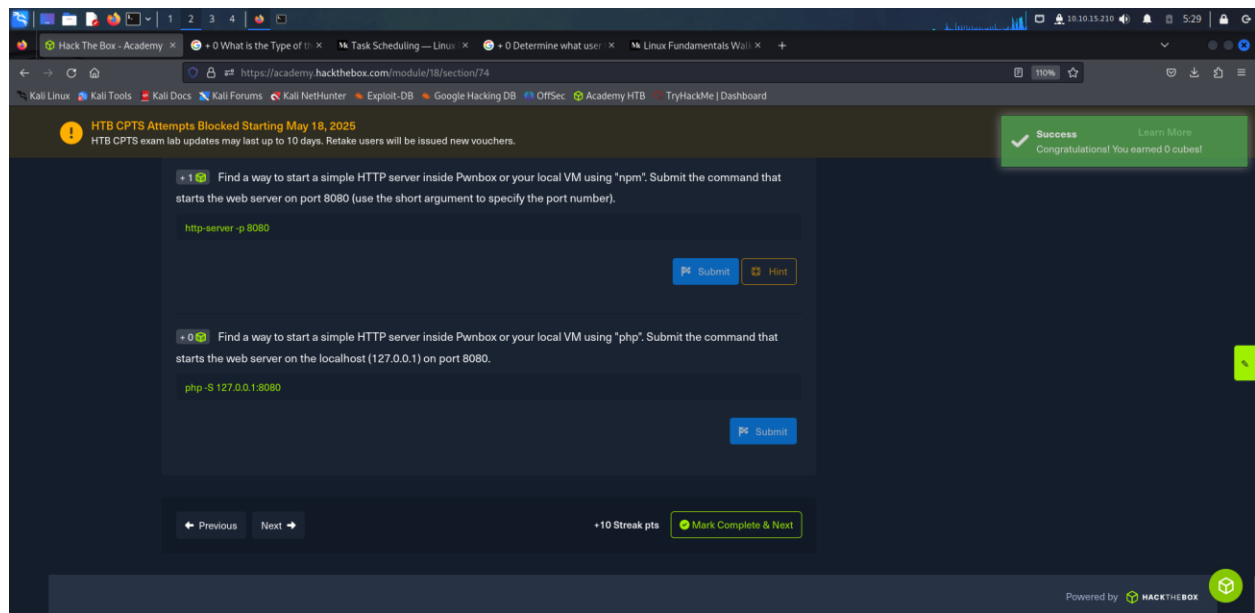


Question: Find a way to start a simple HTTP server inside Pwbox or your local VM using "npm". Submit the command that starts the web server on port 8080 (use the short argument to specify the port number). Answer: `http-server -p 8080`

To start a simple HTTP server on port 8080 on my system, I used the command `http-server -p 8080`

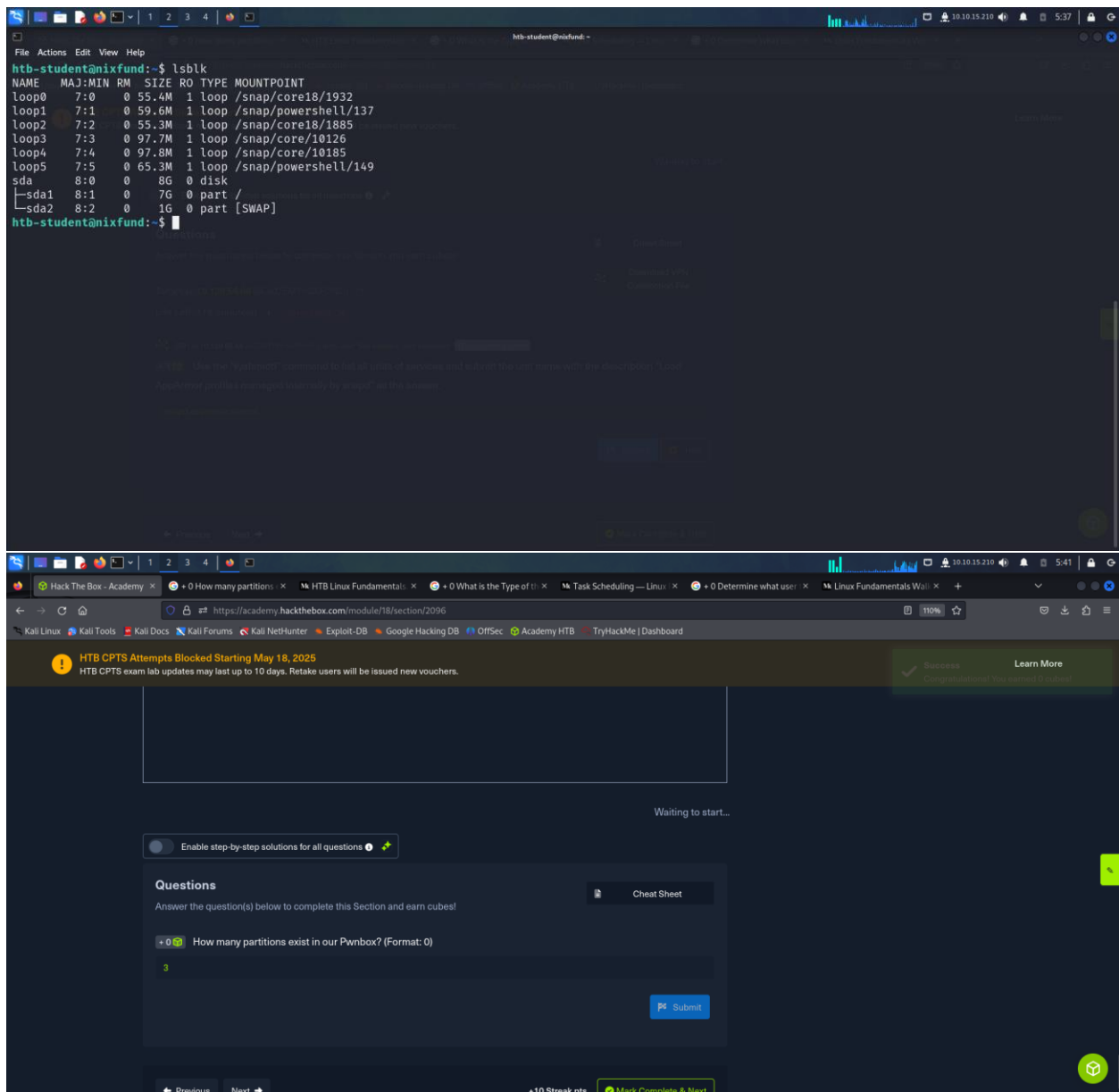
Question: Find a way to start a simple HTTP server on the localhost (127.0.0.1) on port 8080. Answer: `php -S 127.0.0.1:8080`

To start a simple HTTP server on localhost at port 8080 on my system, I used the command `php -S 127.0.0.1:8080`



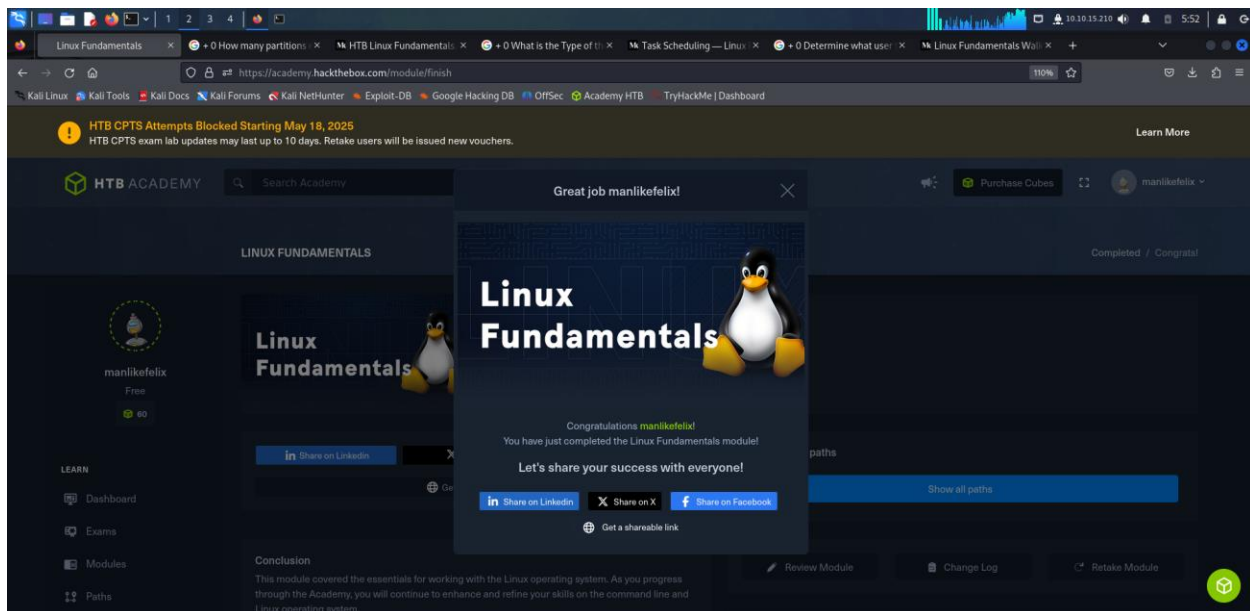
Question: How many partitions exist in our Pwbox? (Format: 0) Answer: 3

To view the block device information on my system, I used the command lsblk. The lsblk command listed all block devices, such as hard drives, partitions, and their mount points, in a tree-like format



3. Shareable Link and Final Screenshot

<https://academy.hackthebox.com/achievement/1919044/18>



4. Social Media Posting

https://www.linkedin.com/uas/login?session_redirect=https%3A%2F%2Fwww.linkedin.com%2FshareArticle%3Furl%3Dhttps%3A%2F%2Facademy.hackthebox.com%2Fachievement%2F1919044%2F18%26title%3DCompleted%2520Linux%2520Fundamentals!%26summary%3DI%2520just%2520completed%2520the%2520module%2520Linux%2520Fundamentals%2520in%2520HTB%2520Academy!%23hackthebox%23htbacademy%23cybersecurity&source=HTBAcademy

<https://x.com/intent/tweet?url=https://academy.hackthebox.com/achievement/1919044/18&text=I just completed module Linux Fundamentals in HTB Academy!&hashtags=hackthebox,htbacademy,cybersecurity>

<https://www.facebook.com/sharer/sharer.php?u=https://academy.hackthebox.com/achievement/1919044/18>

5. Conclusion

Looking back on my journey of learning Linux fundamentals through these exercises, I feel a mix of challenge and accomplishment. Figuring out how to launch a web server with "http-server -p 8080" or "php -S 127.0.0.1:8080" made me realize how accessible and powerful Linux tools can be. Discovering the hardware name "x86_64" and navigating to directories like "/home/htb-student" and "/var/mail/htb-student" helped me get comfortable with the file system and system architecture. Counting the partitions in my Pwnbox as "3" was a practical lesson in resource management that I'll carry forward. Each step of this process has boosted my confidence and sparked my curiosity to explore Linux further, pushing me to take on more complex challenges and apply what I've learned in real-world scenarios down the line.